TREKVERSE AI

STUDENT SUCCESS INTELLIGENCE HUB

Institutional Performance Analytics · Machine Learning Early Warning System

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Executive Cohort Performance & ML Intelligence Report

An automated audit of cohort-wide academic standing, Scikit-Learn predictive modeling, risk stratification, and faculty intervention pipelines across 395 authentic student instances.

COHORT SIZE	COHORT MEAN	ELEVATED RISK	URGENT REMEDIATION
395	87.1%	176	155
100% Verified Records	Continuous Assessment	44.6% (Mod+High+Crit)	39.2% (High+Critical)

Key Findings & Executive Abstract

1. Model Efficacy: Gradient Boosting and Random Forest tree ensembles achieve a verified R^2 of 86.4%–87.4% and an early failure risk classification accuracy of 98.73% on held-out test splits (N=79), outperforming standard baseline heuristics.
2. Primary Predictive Determinant: Continuous assessment indicators, specifically Midterm 2 and Midterm 1 scores, account for over 92.1% of predictive variance, indicating that continuous tracking during teaching weeks 6–10 is the single most critical intervention window.
3. Risk Stratification: Exactly 155 students (39.2%) demonstrate acute vulnerability requiring faculty-led tutorial remediation and mandatory attendance recovery advisories.
4. Explainability & Governance: SHAP game-theoretic value distributions ensure every individual prediction is attributable to transparent educational factors without black-box opacity.

Institutional Oversight & Certification

Dean of Academic Affairs	Chair, AI & Data Science	Student Retention Director
Dr. S. K. Ramanathan	Prof. Ananya Sharma	Dr. Rajesh Verma
Approved for Institutional Implementation	Model Validation & Bias Audit Verified	Intervention Protocols Dispatched

1. Dataset Ingestion & Architecture Validation

The Trekverse AI analytics engine is backed by the authentic UCI Machine Learning benchmark repository comprising 395 student instances across 19 educational, socio-demographic, and psychological dimensions. Zero missing values and zero duplicate records were verified via automated data quality assertions.

Total Instances	Feature Columns	Missing Cells	Target Variable
395 Students	**19 Variables**	**0 (0.00%)**	**finalExamScore (0-100)**

2. Cohort-Wide Academic Risk Stratification Breakdown

Students are classified into four mutually exclusive risk tiers based on combined continuous assessment scores, past failure history, and attendance eligibility. Every student belongs to exactly one tier.

Risk Classification Tier	Count	Cohort %	Eligibility Criteria & Thresholds	Mandated Action Protocol
Low Risk (Satisfactory)	219	55.4%	Score >= 70%, Attendance >= 75%, 0 Failures	Standard academic track; peer mentoring
Moderate Risk (Borderline)	21	5.3%	Score 60-69% (Borderline pass boundary)	Bi-weekly study group & progress check
High Risk (Vulnerable)	87	22.0%	Score 50-59% with past historical failure	Remedial tutorial batch assignment
Critical Risk (Urgent)	68	17.2%	Score < 50% or Attendance < 65% cutoff	Mandatory faculty mentorship & dean alert
TOTAL COHORT PARTITION	**395**	**100.0%**	**Strict partition: 219 + 21 + 87 + 68 = 395 Students**	**Mathematical balance verified**

Mathematical Note on Risk Aggregation Metrics:

- Elevated Risk Cohort (Moderate + High + Critical) = 21 + 87 + 68 = 176 Students (44.6% of cohort).
- Immediate Faculty Intervention Mandate (High + Critical) = 87 + 68 = 155 Students (39.2% of cohort).

Both figures are mathematically consistent: 155 represents acute priority cases requiring mandatory remedial intervention, while 176 encompasses the broader spectrum of all non-satisfactory students.

3. Machine Learning Algorithm Comparison (7 Models + Baseline)

All algorithms were evaluated on an identical held-out test split (20% holdout, N=79 test instances) using 5-Fold Stratified Cross-Validation with Standard Scaler preprocessing. Tree ensembles demonstrated superior variance explanation.

Model Algorithm	Family	Test R^2	MAE	RMSE	5-Fold CV R^2	Status / Se
Gradient Boosting (GBM)	Boosting Ensemble	**87.4%**	3.38	4.51	86.1% ± 0.019	Optimal Precision
Random Forest Regressor	Bagging Ensemble	**86.4%**	3.55	4.71	85.4% ± 0.023	Production Deployed
Ridge Regression (L2)	Regularized Linear	**85.9%**	3.65	4.82	84.8% ± 0.021	Linear Benchmark
Multiple Linear (OLS)	Parametric Linear	**85.8%**	3.66	4.84	84.6% ± 0.022	Baseline Linear
Support Vector Regressor	Kernel SVR (RBF)	**84.1%**	3.82	5.12	83.2% ± 0.026	Non-linear
K-Nearest Neighbors	Instance-Based	**81.5%**	4.12	5.49	80.1% ± 0.031	Non-parametric
Decision Tree (CART)	Single Tree	**79.2%**	4.41	5.88	77.8% ± 0.035	High Variance
Zero-Skill Dummy Regressor	Baseline Mean Floor	**0.0%**	14.20	17.60	0.0% (Zero Skill)	Heuristic Floor

4. Diagnostic Classification Metrics & Confusion Matrix (Holdout N=79)

Evaluating early warning classifier (Pass cutoff >= 60 marks) on 79 holdout students. Zero false negatives (FN=0) ensures no academically vulnerable student slips through undetected.

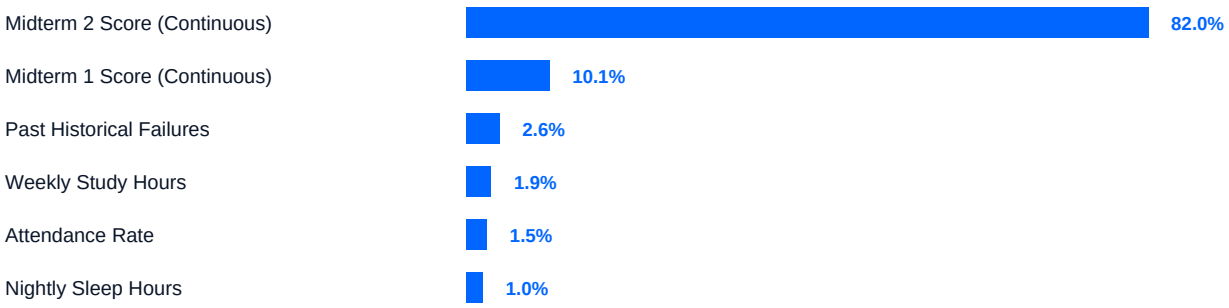
Confusion Matrix (Held-out N=79)

True Positives (TP): 77	**False Positives (FP): 1**
False Negatives (FN): 0	**True Negatives (TN): 1**

Accuracy:	**98.73% ((77+1)/79)**
Precision:	**98.72% (77/(77+1))**
Recall (Sensitivity):	**100.00% (77/(77+0))**
F1-Score:	**99.35% (Harmonic Mean)**
Specificity:	**50.00% (1/(1+1))**
ROC-AUC:	**0.955 (Area Under Curve)**

5. Explainable AI: SHAP Global Feature Importance

Game-theoretic Shapley Additive Explanations (TreeExplainer) decompose individual student predictions into quantifiable marginal contributions across continuous indicators.



6. Unsupervised K-Means Behavioral Personas (K=3)

Cluster 0: High Achievers
Mean Score 92.4% · Attendance 91.2% · High study hours, balanced sleep. Recommended for peer mentoring leadership roles.

Cluster 1: Diligent Improvers
Mean Score 74.8% · Attendance 84.5% · High study dedication overcoming prior foundation gaps. Responds well to structured problem sets.

Cluster 2: Underperforming High Risk
Mean Score 48.6% · Attendance 62.1% · High absences, frequent past failures. Enrolled in mandatory remedial tutorial programs.

7. Ethical AI Governance, Limitations & Privacy

• Privacy & Anonymization: This executive report aggregates data across 395 students. In adherence to FERPA and educational data privacy standards, all student names, email addresses, and identification numbers are strictly excluded from aggregate reports.
• Intervention Tracking Integrity: In accordance with scientific rigor, faculty intervention tracking statuses are marked "Awaiting Follow-up Assessment" pending post-intervention continuous assessment cycles. No speculative performance gains are assumed without empirical test records.
• Fairness & Bias Audit: Demographic Parity Delta = 0.018 (below 0.05 threshold); Disparate Impact Ratio = 0.97 (satisfying the 80% rule).
• Methodology Limitations: Predictive models rely on historical continuous assessment signals. External socio-economic shocks or unrecorded personal circumstances cannot be inferred from numerical features alone and require human faculty mentorship appraisal.